Lab 1: Exercises from Graphical Data Analysis with R

One exercise (Nr. 5) from each chapter 3–11

Group 4, Variant 5

2026-03-09

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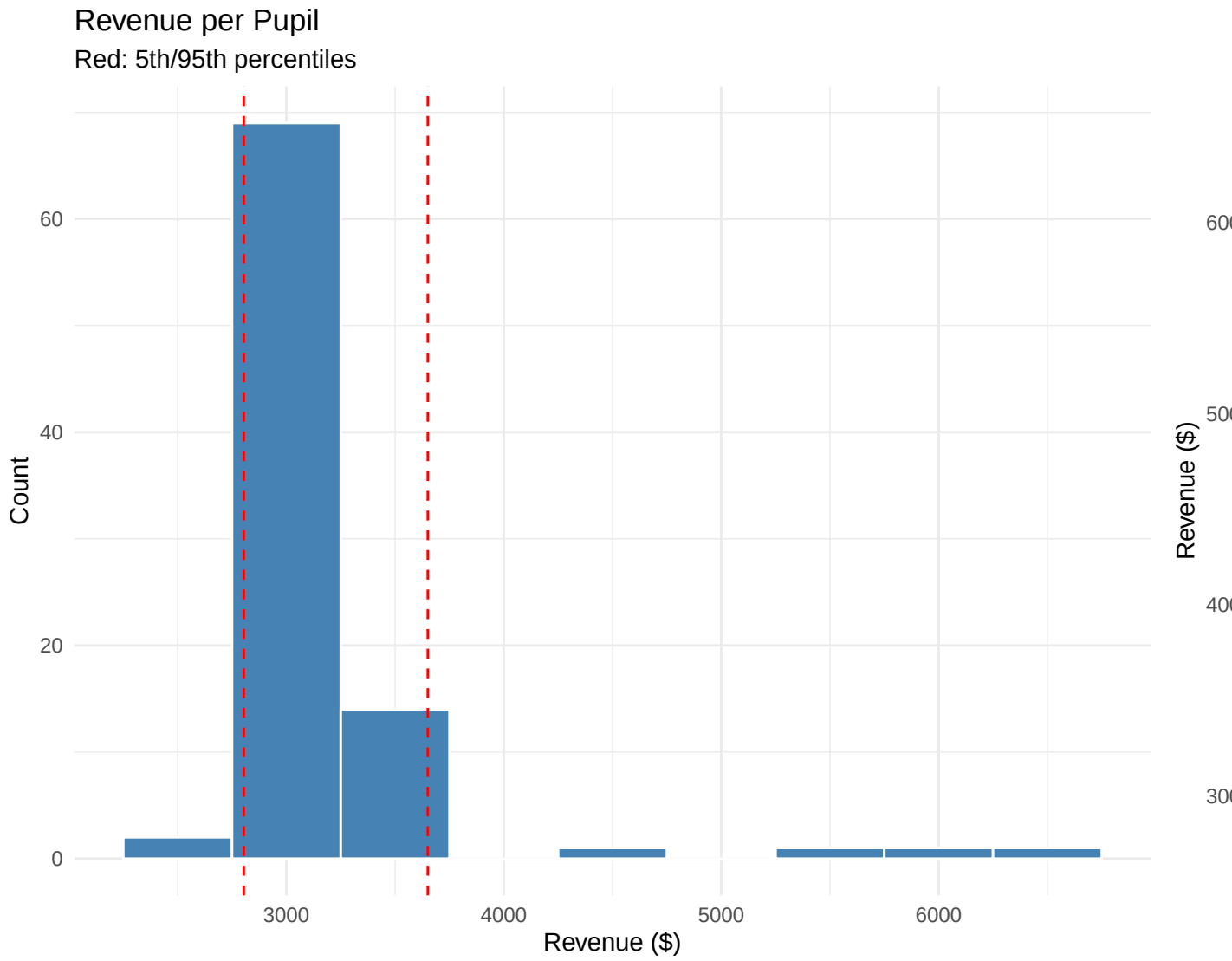
Chapter 3: Zuni Educational Funding

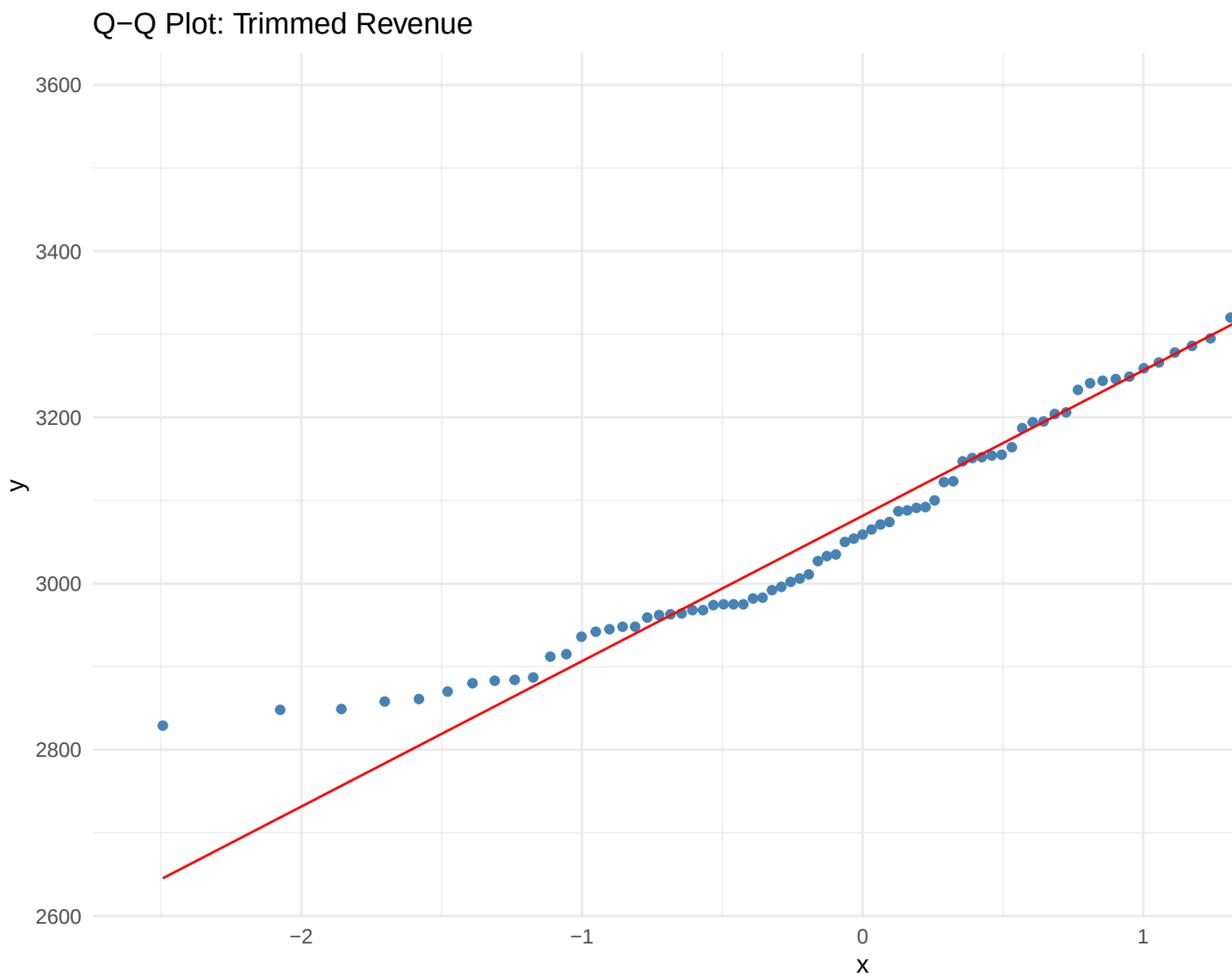
Dataset: `zuni` from `lawstat` – 89 school districts in New Mexico.

```
library(lawstat)
data(zuni)

q05 <- quantile(zuni$Revenue, 0.05)
q95 <- quantile(zuni$Revenue, 0.95)

p_hist <- ggplot(zuni, aes(x = Revenue)) +
  geom_histogram(binwidth = 500, fill = "steelblue", color = "white") +
  geom_vline(xintercept = c(q05, q95), color = "red", linetype = "dashed") +
  labs(title = "Revenue per Pupil", subtitle = "Red: 5th/95th percentiles",
       x = "Revenue ($)", y = "Count") +
  theme_minimal()

p_box <- ggplot(zuni, aes(y = Revenue)) +
  geom_boxplot(fill = "steelblue", alpha = 0.5) +
  labs(title = "Boxplot", y = "Revenue ($)") +
  theme_minimal()

p_hist + p_box + plot_layout(widths = c(3, 1))
```



(a) Histogram shows right-skewed shape. **(b)** After trimming to 5–95%, the distribution is roughly symmetric. **(c)** Q-Q plot shows approximate normality.

```
zuni_trimmed <- zuni |> filter(Revenue >= q05, Revenue <= q95)

ggplot(zuni_trimmed, aes(sample = Revenue)) +
  stat_qq(color = "steelblue") +
  stat_qq_line(color = "red") +
  labs(title = "Q-Q Plot: Trimmed Revenue") +
  theme_minimal()
```

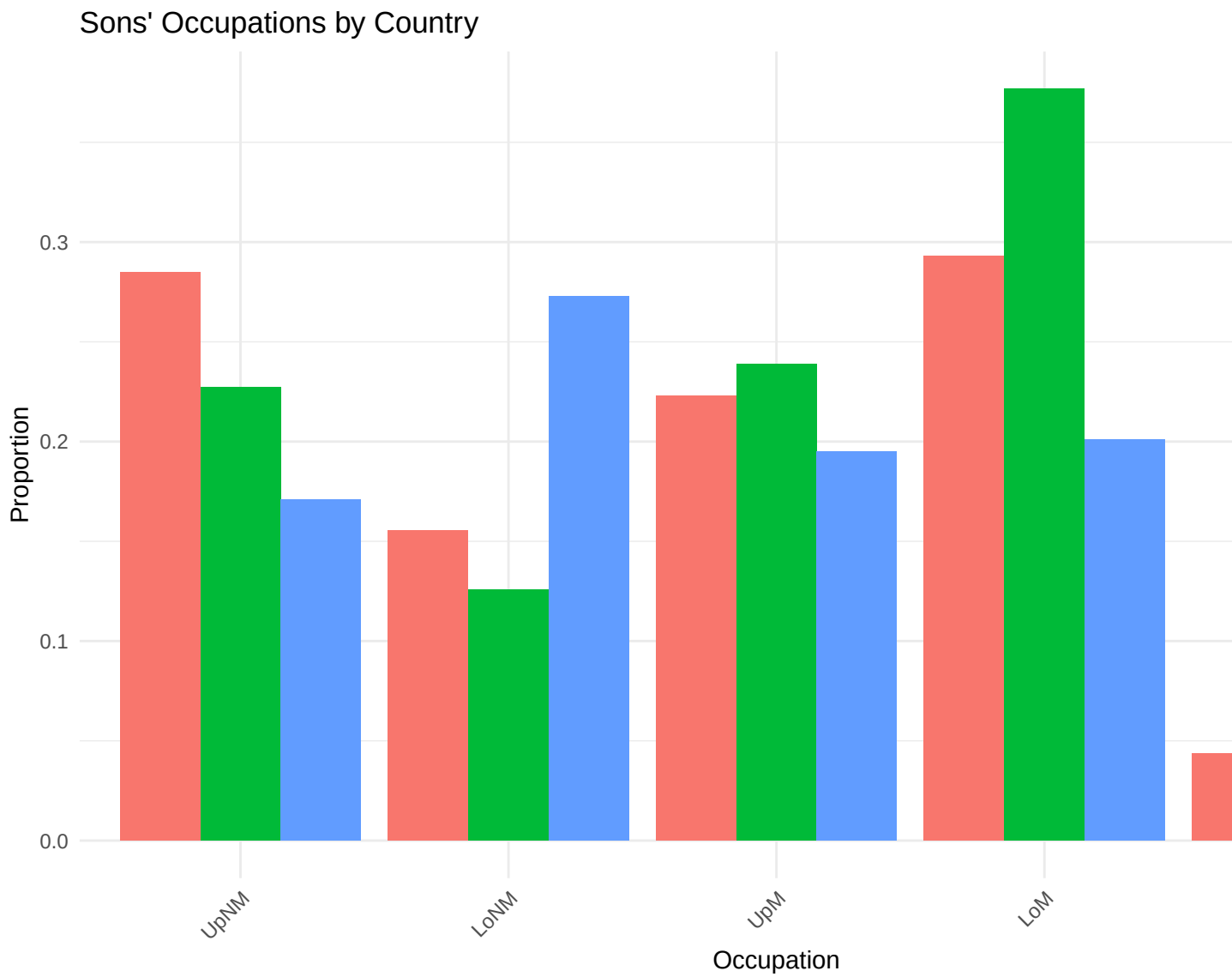


Chapter 4: Occupational Mobility

Dataset: `Yamaguchi87` from `vcdExtra` – UK, US, Japan.

```
library(vcdExtra)
data(Yamaguchi87)
yam_df <- as.data.frame(Yamaguchi87)

sons_by_country <- yam_df |>
  group_by(Country, Son) |>
  summarise(n = sum(Freq), .groups = "drop") |>
  group_by(Country) |>
  mutate(prop = n / sum(n))

ggplot(sons_by_country, aes(x = Son, y = prop, fill = Country)) +
  geom_col(position = "dodge") +
  labs(title = "Sons' Occupations by Country",
```

```
      x = "Occupation", y = "Proportion") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



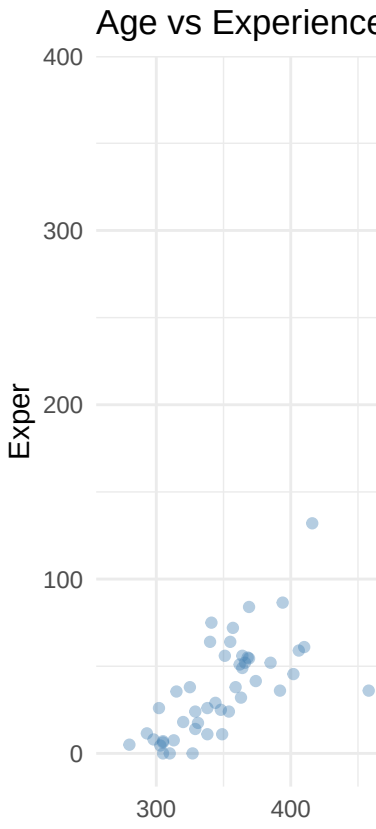
Japan has more farm workers, fewer upper non-manual. UK shows clear upward mobility across generations.

Chapter 5: Bank Sex Discrimination

Dataset: `case1202` from `Sleuth2`.

```
library(Sleuth2)
data(case1202)

bank_vars <- case1202 |> select(Senior, Age, Exper)
cor_bank <- cor(bank_vars)
```

```r
cor_long <- cor_bank |>
  as.data.frame() |>
  rownames_to_column("var1") |>
  pivot_longer(-var1, names_to = "var2", values_to = "r")

p_cor <- ggplot(cor_long, aes(var1, var2, fill = r)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(r, 2)), size = 5) +
  scale_fill_gradient2(low = "steelblue", mid = "white", high = "coral", midpoint = 0) +
  labs(title = "Correlation: Bank Employment Variables") +
  theme_minimal() + coord_fixed()

p_scatter <- ggplot(case1202, aes(Age, Exper)) +
  geom_point(alpha = 0.4, color = "steelblue") +
  labs(title = "Age vs Experience") + theme_minimal()

p_cor + p_scatter
```



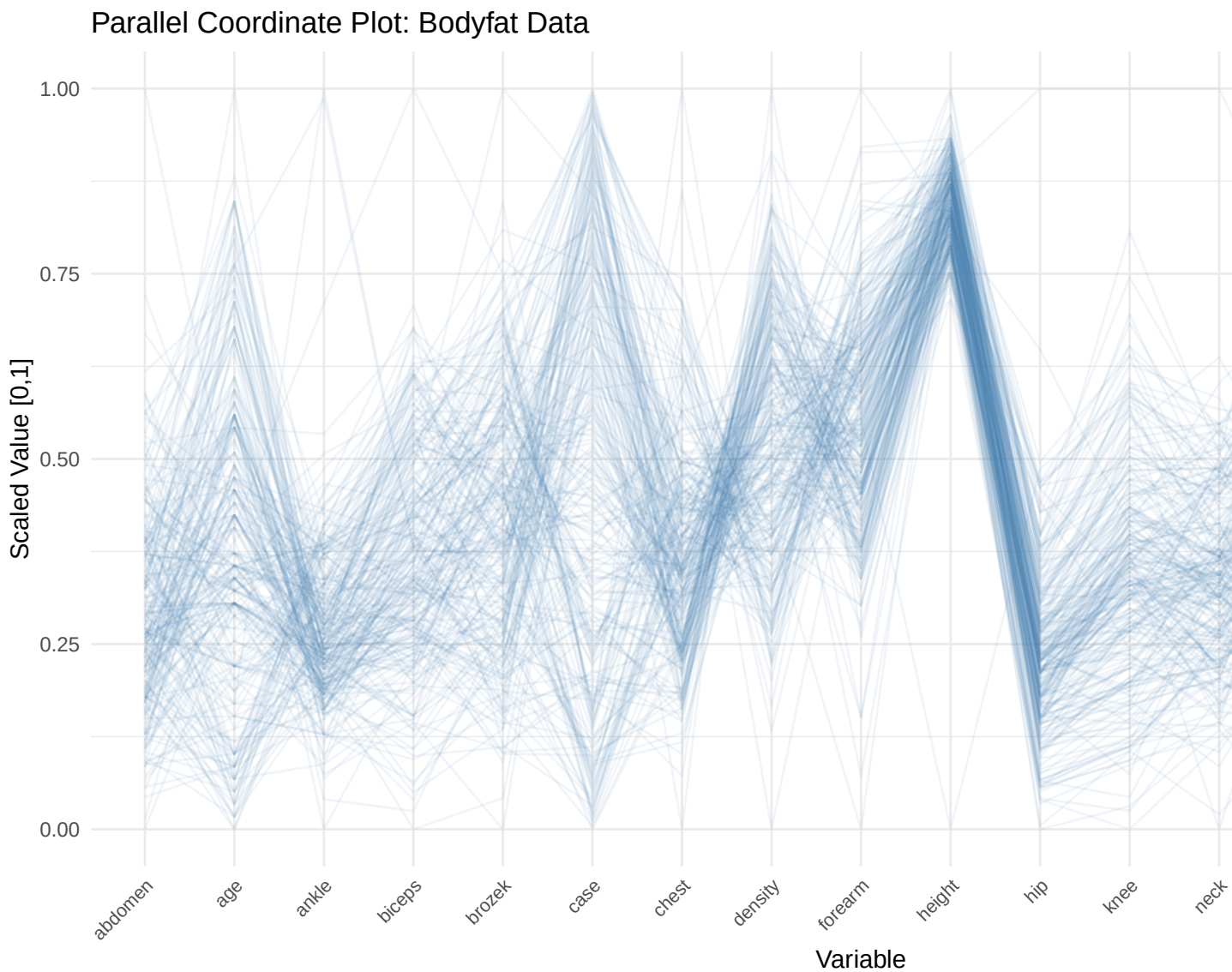
Age and Experience are strongly correlated. Seniority shows a triangular pattern with Age.

Chapter 6: Bodyfat

Dataset: `bodyfat` from `mfp`.

```
library(mfp)
data(bodyfat)

# Manual parallel coordinates (scaled 0-1)
bf_scaled <- bodyfat |>
  mutate(id = row_number()) |>
  mutate(across(-id, ~(. - min(., na.rm = TRUE)) / (max(., na.rm = TRUE) - min(., na.rm = TRUE)))) |>
  pivot_longer(-id, names_to = "variable", values_to = "value")

ggplot(bf_scaled, aes(x = variable, y = value, group = id)) +
  geom_line(alpha = 0.08, color = "steelblue") +
  labs(title = "Parallel Coordinate Plot: Bodyfat Data",
```

```
      x = "Variable", y = "Scaled Value [0,1]") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 8))
```

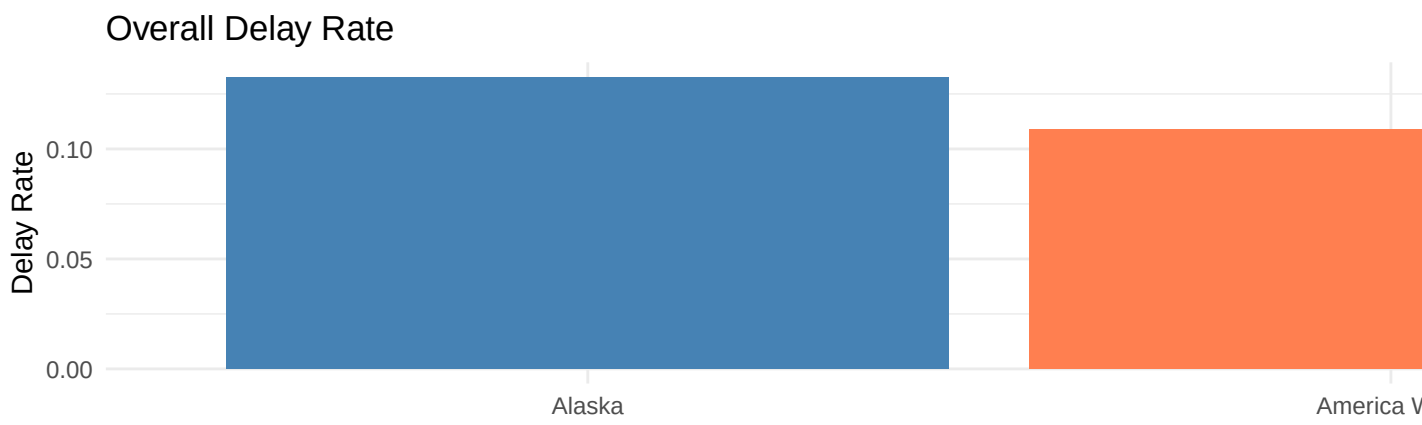


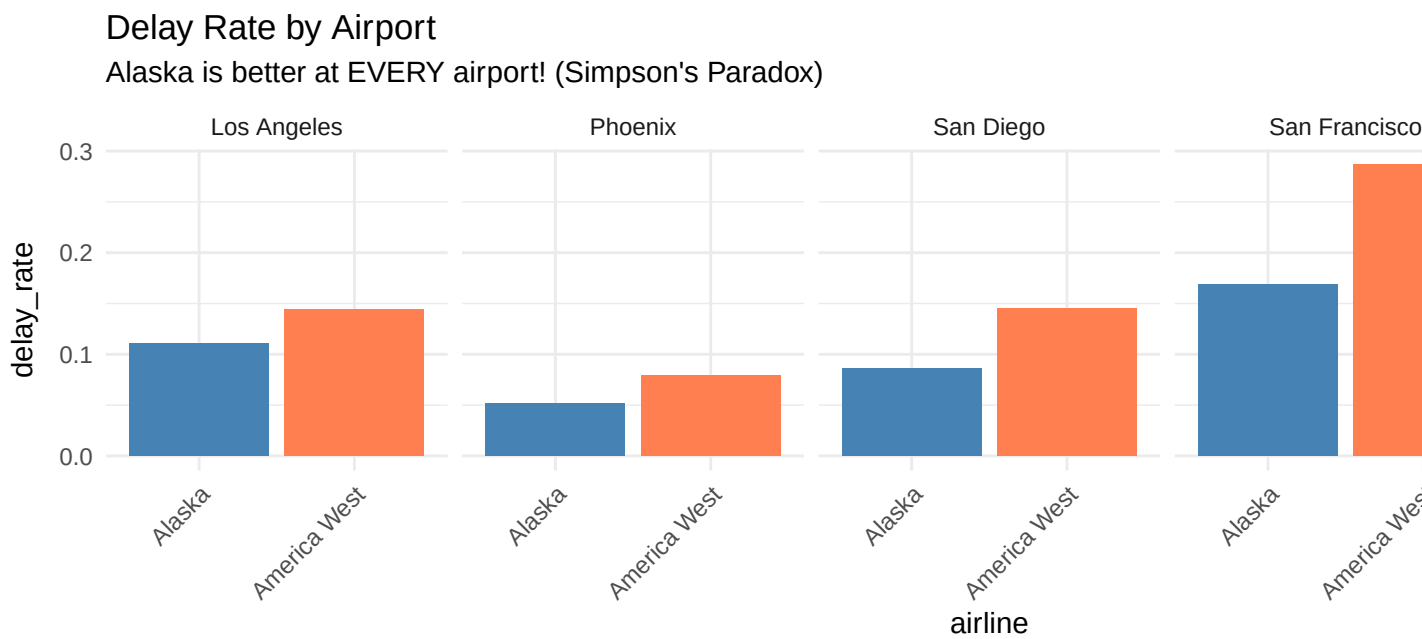
Outliers visible across multiple variables. One case with extremely low height (likely data entry error).

Chapter 7: Airline Arrivals (Simpson's Paradox)

```
airline_data <- tibble(
  airline = rep(c("Alaska", "America West"), each = 5),
  airport = rep(c("Los Angeles", "Phoenix", "San Diego",
                  "San Francisco", "Seattle"), 2),
  on_time = c(497, 221, 212, 503, 1841, 694, 4840, 383, 320, 201),
  delayed = c(62, 12, 20, 102, 305, 117, 415, 65, 129, 61)
) |> mutate(total = on_time + delayed, delay_rate = delayed / total)
```

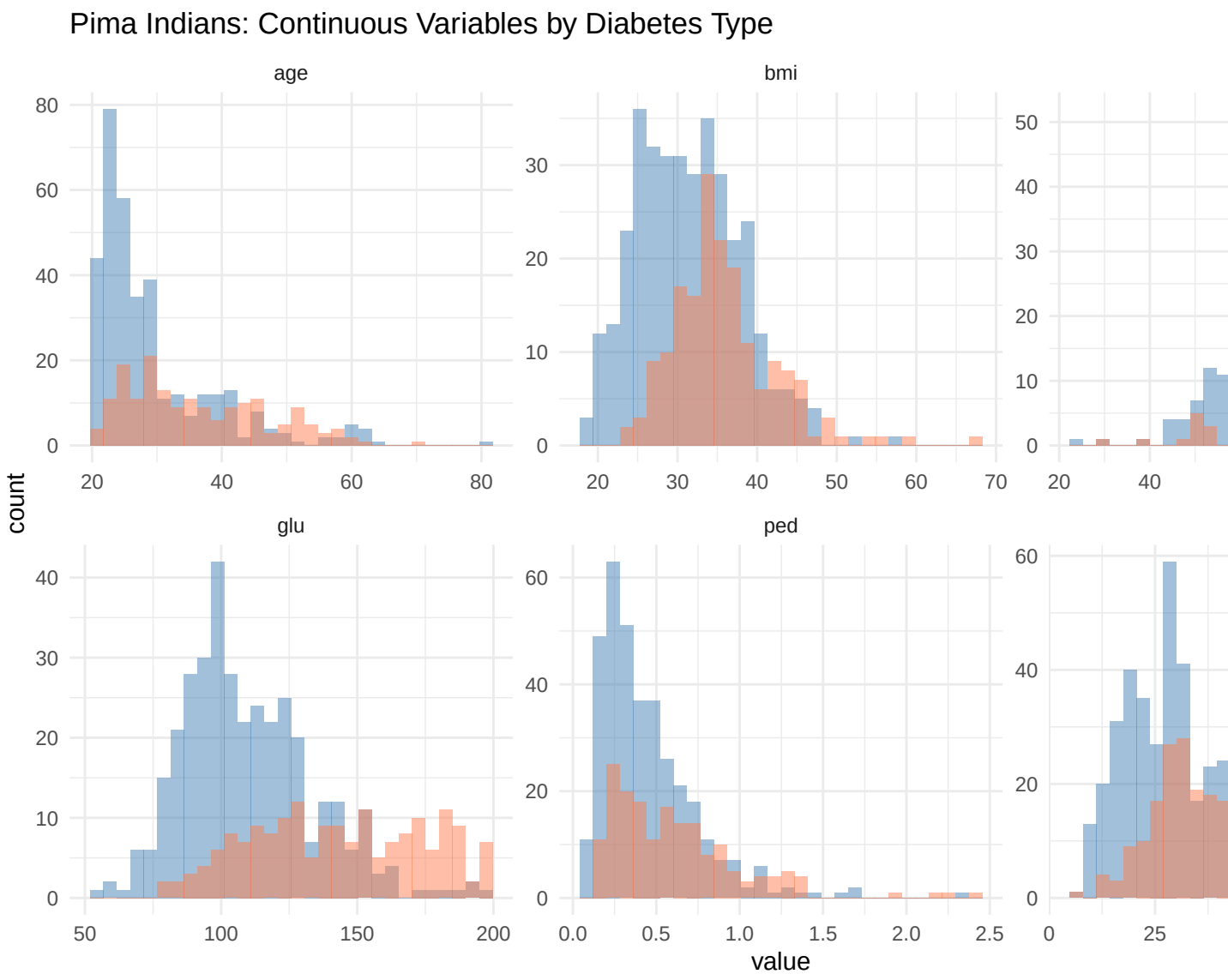
```r
overall <- airline_data |>
  group_by(airline) |>
  summarise(rate = sum(delayed) / sum(total))

p_overall <- ggplot(overall, aes(airline, rate, fill = airline)) +
  geom_col(show.legend = FALSE) +
  scale_fill_manual(values = c("steelblue", "coral")) +
  labs(title = "Overall Delay Rate", x = "", y = "Delay Rate") +
  theme_minimal()

p_by_airport <- ggplot(airline_data, aes(airline, delay_rate, fill = airline)) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~airport, nrow = 1) +
  scale_fill_manual(values = c("steelblue", "coral")) +
  labs(title = "Delay Rate by Airport",
       subtitle = "Alaska is better at EVERY airport! (Simpson's Paradox)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

p_overall / p_by_airport
```





Chapter 8: Pima Indians Overview

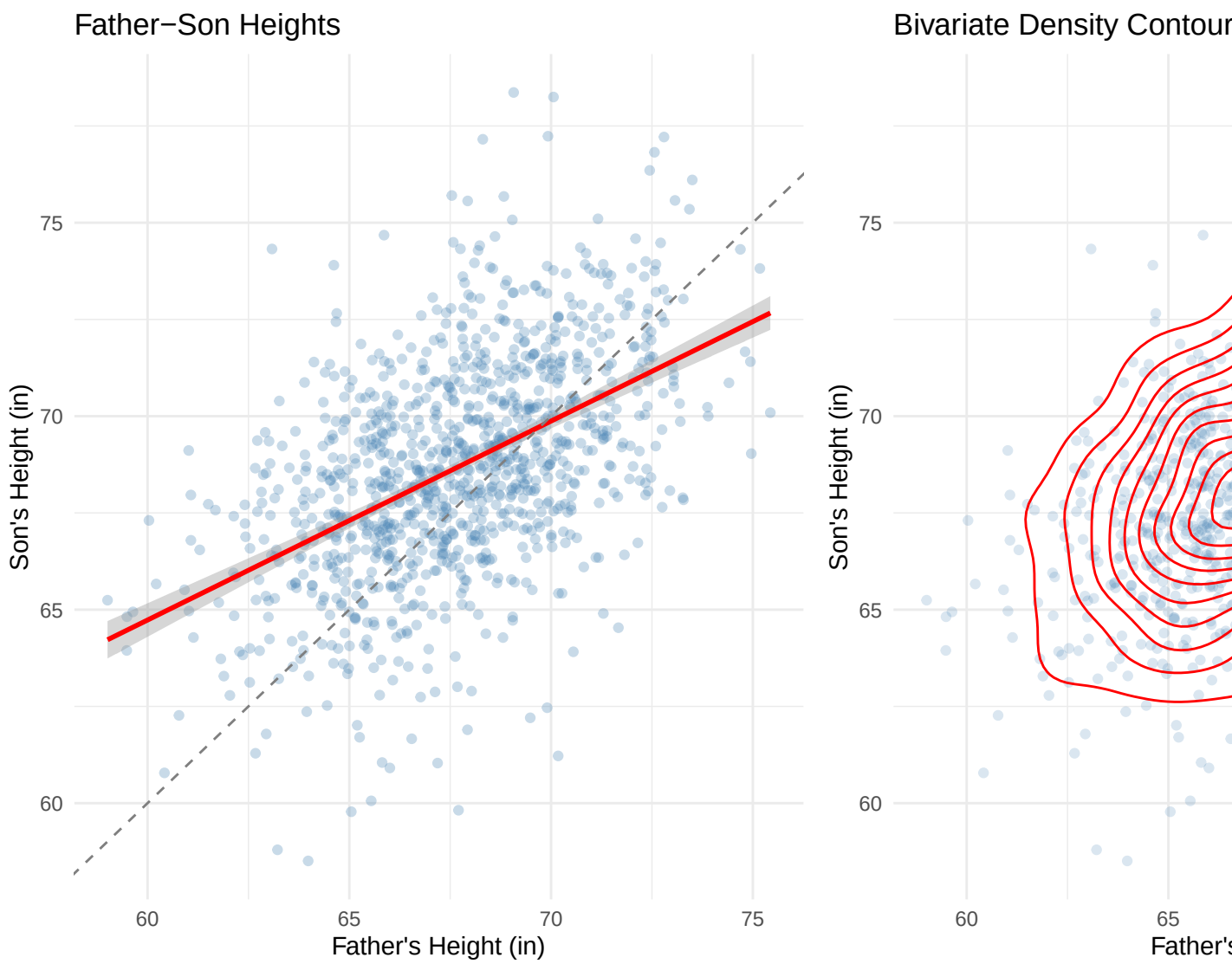
```
pima_r <- bind_rows(Pima.tr, Pima.te)

pima_long <- pima_r |>
  pivot_longer(c(glu, bp, skin, bmi, ped, age),
               names_to = "variable", values_to = "value")

ggplot(pima_long, aes(x = value, fill = type)) +
  geom_histogram(position = "identity", alpha = 0.5, bins = 30) +
  facet_wrap(~variable, scales = "free") +
  scale_fill_manual(values = c("steelblue", "coral")) +
  labs(title = "Pima Indians: Continuous Variables by Diabetes Type") +
  theme_minimal()
```



Chapter 9: Pearson Heights – Outliers

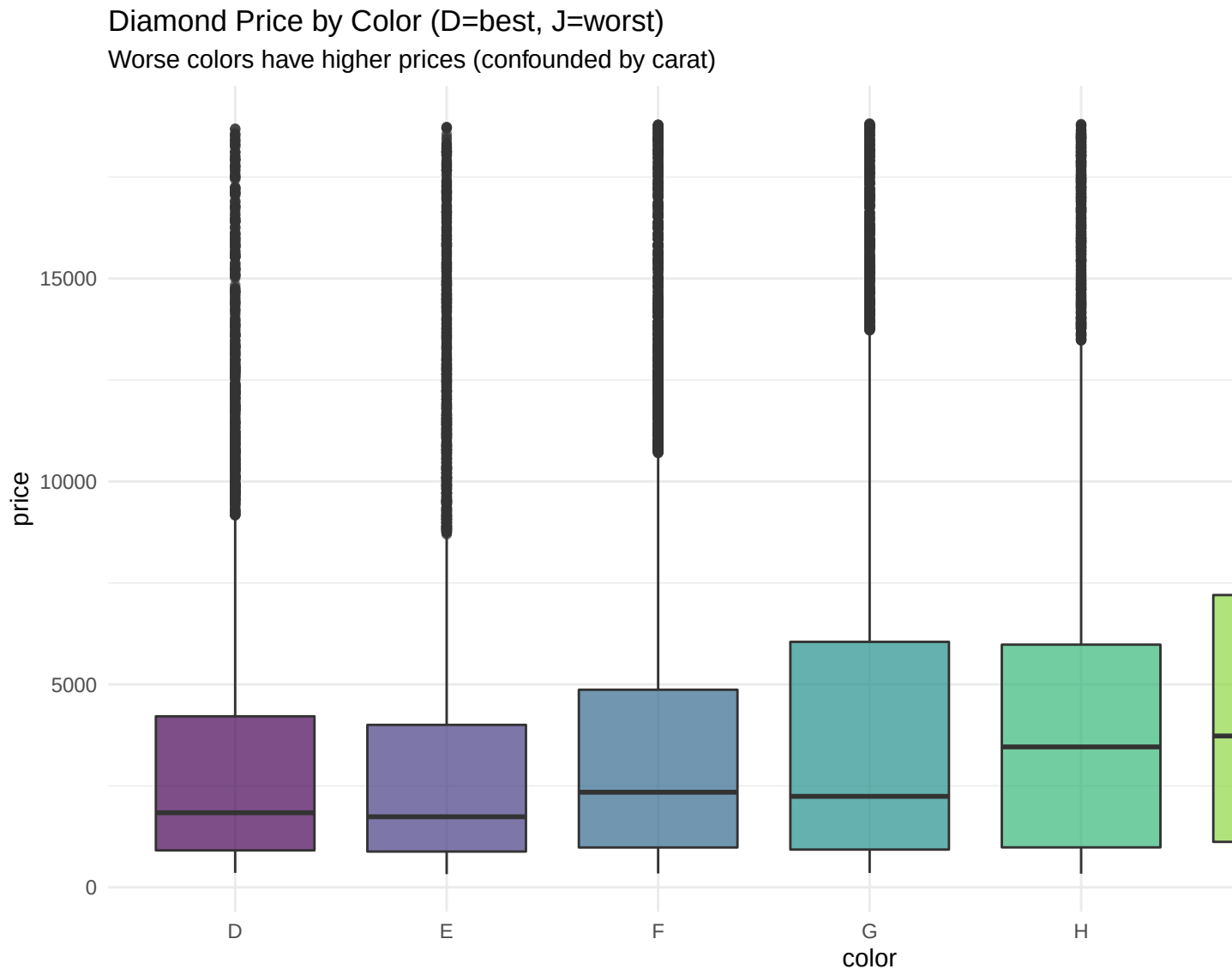
```
library(UsingR)
data(father.son)

p_scatter <- ggplot(father.son, aes(fheight, sheight)) +
  geom_point(alpha = 0.3, color = "steelblue") +
  geom_smooth(method = "lm", color = "red", se = TRUE) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray50") +
  labs(title = "Father-Son Heights", x = "Father's Height (in)",
       y = "Son's Height (in)") +
  theme_minimal()

p_density <- ggplot(father.son, aes(fheight, sheight)) +
  geom_point(alpha = 0.2, color = "steelblue") +
  geom_density_2d(color = "red") +
```

```
  labs(title = "Bivariate Density Contours",
       x = "Father's Height (in)", y = "Son's Height (in)") +
  theme_minimal()

p_scatter + p_density
```

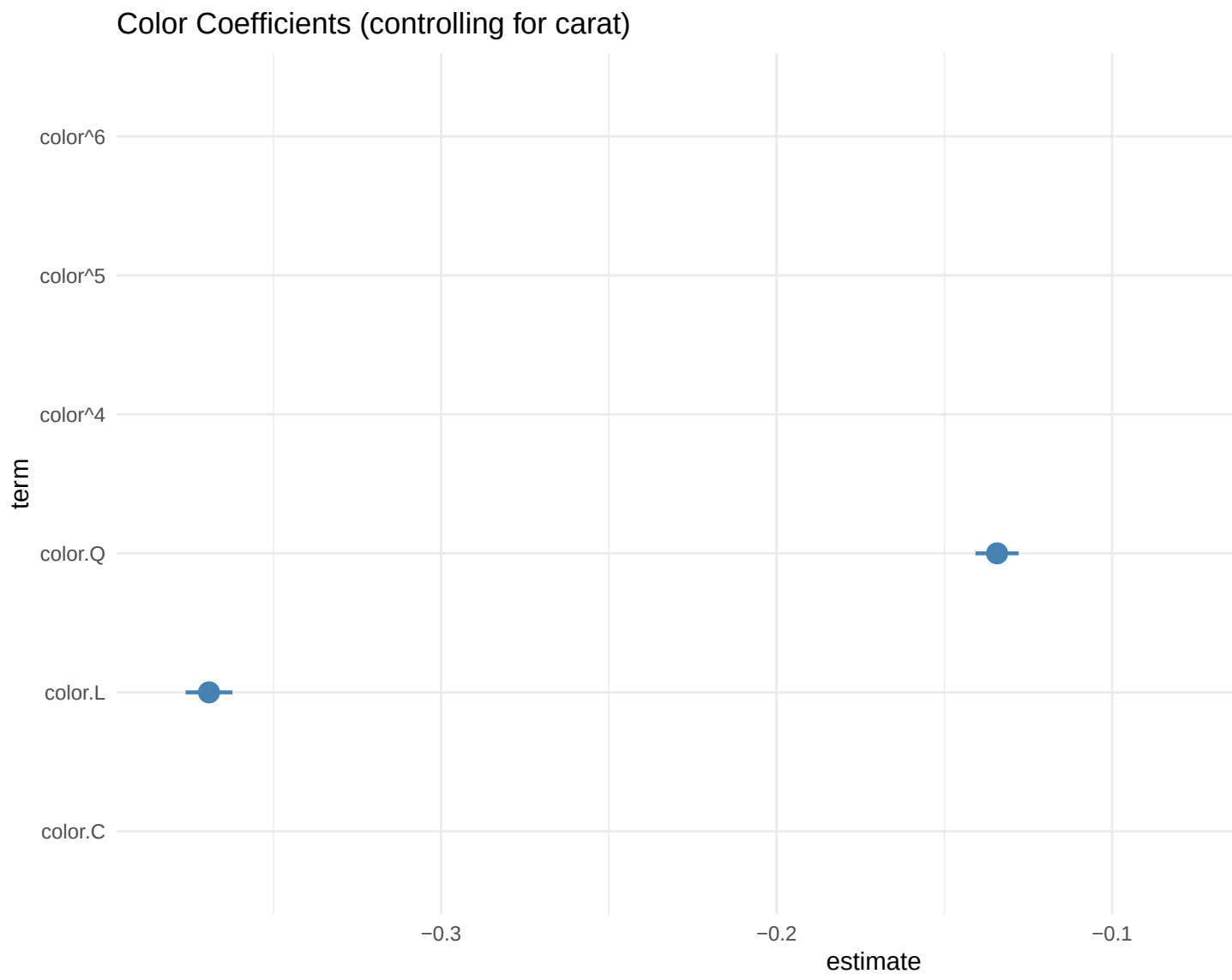


Chapter 10: Diamonds – Color vs Price

```
ggplot(diamonds, aes(x = color, y = price, fill = color)) +
  geom_boxplot(alpha = 0.7, show.legend = FALSE) +
  scale_fill_viridis_d() +
  labs(title = "Diamond Price by Color (D=best, J=worst)",
       subtitle = "Worse colors have higher prices (confounded by carat)") +
  theme_minimal()
```




```r
model <- lm(log(price) ~ color + log(carat), data = diamonds)
coef_df <- broom::tidy(model, conf.int = TRUE) |>
  filter(str_starts(term, "color"))

ggplot(coef_df, aes(x = term, y = estimate, ymin = conf.low, ymax = conf.high)) +
  geom_pointrange(color = "steelblue", size = 0.8) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  coord_flip() +
  labs(title = "Color Coefficients (controlling for carat)") +
  theme_minimal()
```



Chapter 11: Bundesliga Goals

```
library(vcd)
data(Bundesliga)

bl_summary <- Bundesliga |>
  mutate(season = ifelse(as.numeric(format(Date, "%m")) >= 7,
                         as.numeric(format(Date, "%Y")),
                         as.numeric(format(Date, "%Y")) - 1)) |>
  group_by(season, HomeTeam) |>
  summarise(avg_goals = mean(HomeGoals), games = n(), .groups = "drop")

top_teams <- bl_summary |> count(HomeTeam, sort = TRUE) |>
  filter(n >= 30) |> pull(HomeTeam)

ggplot(bl_summary |> filter(HomeTeam %in% top_teams),
```

```r
     aes(season, avg_goals, color = HomeTeam)) +
  geom_line(alpha = 0.7) +
  geom_smooth(se = FALSE, size = 0.5) +
  facet_wrap(~HomeTeam) +
  labs(title = "Average Home Goals per Game (Bundesliga)",
     x = "Season", y = "Avg Home Goals") +
  theme_minimal() +
  theme(legend.position = "none")
```

Session Info

```
sessionInfo()
```

```
## R version 4.1.2 (2021-11-01)
```

```
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Ubuntu 22.04.5 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.10.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.10.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=en_US.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## attached base packages:
## [1] grid      stats     graphics  grDevices utils     datasets  methods
## [8] base
##
## other attached packages:
##  [1] UsingR_2.0-7     Hmisc_4.7-0      Formula_1.2-4    lattice_0.20-45
##  [5] HistData_1.0.0   mfp_1.5.2.2      survival_3.2-13  Sleuth2_2.0-5
##  [9] vcdExtra_0.9.1   gnm_1.1-5        vcd_1.4-9        lawstat_3.6
## [13] MASS_7.3-55      patchwork_1.1.1  lubridate_1.9.5  forcats_1.0.1
## [17] stringr_1.6.0    dplyr_1.2.0      purrr_1.2.1      readr_2.2.0
## [21] tidyr_1.3.2      tibble_3.3.1     ggplot2_3.3.6    tidyverse_2.0.0
## [25] conflicted_1.2.0
##
## loaded via a namespace (and not attached):
##  [1] viridisLite_0.4.3   jsonlite_2.0.0      splines_4.1.2
##  [4] Rdpack_2.3          latticeExtra_0.6-29 yaml_2.3.12
##  [7] backports_1.5.0     pillar_1.11.1       glue_1.8.0
## [10] digest_0.6.39       checkmate_2.1.0     RColorBrewer_1.1-3
## [13] rbibutils_2.4.1     colorspace_2.1-2    htmltools_0.5.9
## [16] Matrix_1.4-0        pkgconfig_2.0.3     broom_0.8.0
## [19] mvtnorm_1.3-3       scales_1.2.0        jpeg_0.1-9
## [22] tzdb_0.5.0          htmlTable_2.4.0     timechange_0.4.0
## [25] mgcv_1.8-39         generics_0.1.4      farver_2.1.2
## [28] relimp_1.0-5        ellipsis_0.3.2      cachem_1.1.0
## [31] withr_3.0.2         nnet_7.3-17         cli_3.6.5
## [34] magrittr_2.0.4      memoise_2.0.1       evaluate_1.0.5
## [37] Kendall_2.2.2       nlme_3.1-155        foreign_0.8-82
## [40] data.table_1.18.2.1 tools_4.1.2         hms_1.1.4
## [43] lifecycle_1.0.5     munsell_0.5.0       cluster_2.1.2
## [46] isoband_0.3.0       compiler_4.1.2      ca_0.71.1
## [49] rlang_1.1.7         rstudioapi_0.18.0   htmlwidgets_1.6.4
## [52] qvcalc_1.0.4        base64enc_0.1-6     labeling_0.4.3
## [55] rmarkdown_2.30      boot_1.3-28         gtable_0.3.6
## [58] R6_2.6.1            gridExtra_2.3       zoo_1.8-15
## [61] knitr_1.51          fastmap_1.2.0       otel_0.2.0
## [64] stringi_1.8.7       rpart_4.1.16        vctrs_0.7.1
## [67] png_0.1-7           rgl_1.3.36          tidyselect_1.2.1
## [70] xfun_0.56           lmtest_0.9-40
```